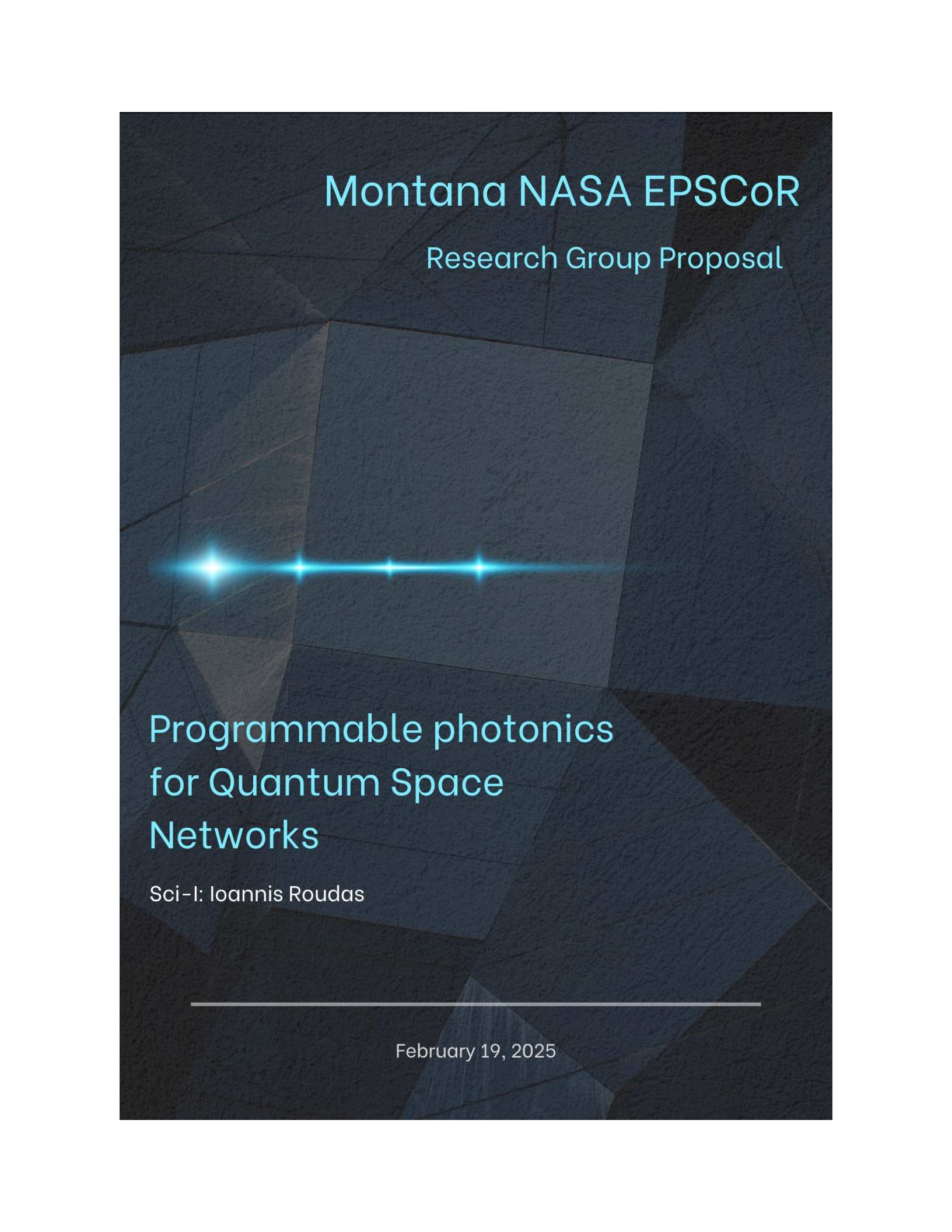




Montana NASA EPSCoR

Research Group Proposal

Programmable photonics for Quantum Space Networks

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February 19, 2025

Abstract

Programmable photonics for Quantum Space Networks

NASA and other space agencies, as well as private companies, are developing free-space optical (FSO) communication systems to meet the growing demand for high-speed connectivity in space missions. These systems will connect space probes, satellites, aircraft, UAVs, and ground stations into a space internet of things that can host both classical and quantum applications.

When laser beams travel through Earth's atmosphere, they refract, diffract, and scatter due to atmospheric turbulence—random temperature-induced variations in the refractive index of air. These effects distort the phase front of the received optical wave, leading to intensity fluctuations (scintillation), beam wandering, and signal fading at the receiver. This lack of reliability on behalf of FSO links is a major obstacle to the widespread adoption of optical communications for space missions.

To overcome this impairment, adaptive optical systems can be used to continuously correct phase distortions in real-time, ensuring stable and efficient laser communications for classical and quantum networks. Conventional adaptive optics in astronomical telescopes rely on slow, bulky, and expensive deformable mirrors. The challenge for space optical communications is to develop compact, real-time adaptive optical equalizers that correct quickly turbulence-induced distortions without excessive weight, power consumption, or complexity. Photonic integration, which miniaturizes thousands of optical components onto a single chip, addresses these challenges while reducing costs through mass production.

This project focuses on programmable photonic integrated circuits (PICs), analogous to electronic field-programmable gate arrays (FPGAs), which can be dynamically reconfigured to perform different functions. In collaboration with the Optics and Photonics Group at NASA's Glenn Research Center (GRC), a multidisciplinary team at Montana State University (MSU) will design, simulate, and experimentally characterize programmable PIC architectures for optical signal processing in future quantum space networks.

The project consists of four key tasks. First, the MSU team will design and simulate scalable adaptive equalizer architectures based on Mach-Zehnder interferometer meshes and phase shifters for integration into programmable PICs. Second, the MSU team will build and characterize a 2×2 tabletop adaptive equalizer using free-space optical components to evaluate its performance under artificial turbulence conditions. For this purpose, the MSU team will develop FPGA-based signal processing to be used for the control of the equalizer. In the third task, MSU will explore the use of programmable PICs for interconnecting quantum memories based on neutral atom arrays. Finally, the NASA GRC team will fabricate and demonstrate working prototype photonic chips, providing realistic specifications and technical support for the first three tasks.

The project builds on MSU's quantum technology infrastructure, including the Applied Quantum CORE, MonArk NSF Quantum Foundry, and Montana Microfabrication Facility (MMF), reinforcing MSU's strategy to invest in classical and quantum photonics technologies and aligning with Montana's growing optics and photonics industry.

NASA EPSCoR funding is critical as it enables a strategic partnership between MSU and NASA GRC's Optics and Photonics Group. This collaboration advances programmable photonic integrated circuits (PICs) for free-

space optical (FSO) communications, aligning with NASA's goals in optical communications and quantum networking. Upon project completion, MSU's ability to secure extramural funding for programmable PICs will significantly improve. Additionally, it will foster workforce development in Montana, training students and researchers in high-tech fields essential to NASA's mission.

Keywords: Free-space optical communications, quantum networking, silicon photonics, programmable photonic integrated circuits.

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1. Scientific/Technical Plan

1.1 Background

Developing high-capacity, reliable communication systems is essential for space exploration [1]. As missions become more advanced, the need for high-data-rate deep-space networks is increasing. While radio frequency (RF) communications have historically supported NASA's space program, they lack the capacity required for future missions. To address this, NASA is transitioning to free-space optical (FSO) communications, which offer higher data rates with minimal lower size, weight, and power (SWaP) [2], [3]. Other salient features include spectrum availability, low cost and easy deployment, high security, and resilience in extreme environments.

Previous and current NASA missions using lasercom

NASA has demonstrated FSO technologies in several recent missions, e.g., the 2013 Lunar Laser Communication Demonstration (LLCD) [4], the 2021 Laser Communications Relay Demonstration (LCRD) [5], and the TeraByte InfraRed Delivery (TBIRD) program with a 200-Gbps link from a CubeSat [6]. The Deep Space Optical Communications (DSOC) experiment [7] will test high-bandwidth laser links from Mars orbit, and Artemis II (2025) will evaluate laser communication from the Orion spacecraft to Earth at up to 260 Mbps [8].

Optical intersatellite links (ISLs): Starlink and other government/corporate programs

Mega-constellations of satellites comprising hundreds and even thousands of satellites are designed by multiple countries and private companies. These constellations use optical intersatellite links (ISLs) [9] to create a high-speed, free-space optical mesh network in the sky, similar to the fiber-optic internet backbone on Earth. SpaceX's Starlink [10], Eutelsat's OneWeb, and Telesat's Lightspeed are operational constellations with optical ISLs while similar satellite networks are under development worldwide, e.g., the European Space Agency's HyDRON, Amazon's Project Kuiper, and China's Qianfan constellation.

Quantum optical satellites

Future internet architectures will integrate high-fidelity quantum entanglement distribution alongside traditional broadband services. Key recent quantum optical space experiments include China's Micius satellite (2017) for Quantum Key Distribution (QKD) [11] and NASA JPL's SEAQUE project (2024) [12], that will test the feasibility of generating and maintaining entangled photons in space outside the International Space Station. These efforts pave the way for secure, long-distance quantum networking in space.

Programmable photonic integrated circuits (PICs)

Future FSO space networks will comprise thousands of optically interconnected satellites. Miniaturized photonic integrated circuits (PICs) will significantly reduce the cost and complexity of each space terminal, facilitating scalable deployment. Currently, silicon photonics and lithium niobate platforms dominate PIC development. Silicon photonics leverage CMOS-based fabrication, allowing low-cost, high-volume production and co-integration with electronics.

Programmable photonics enable dynamic control of photonic chips—akin to how Field Programmable Gate Arrays (FPGAs) revolutionized electronics—and have applications in quantum processors, optical co-processors, and neural networks. PIC-based adaptive optics provide a lightweight, scalable alternative to traditional bulky deformable mirrors, making them ideal for next-generation space-based communications supporting classical and quantum signals. NASA Glenn Research Center (GRC) is advancing programmable PIC-based coherent combining.

1.2 Project description

The proposed project will explore the use of PICs for FSO communication links. In collaboration with the Optics and Photonics Branch of the GRC, we aim to design, fabricate, and experimentally characterize Mach-Zehnder interferometer (MZI)-based programmable PICs for classical and quantum functionalities.

Network architecture

Fig. 1(a) is a conceptual representation of a global FSO network that interconnects satellites, space probes, unmanned aerial vehicles (UAVs), and ground stations to form a space internet of things (IoT) capable of supporting classical and quantum applications. This infrastructure enables high-speed data transfer across space missions.



Fig. 1 (a) Envisioned FSO mesh network for high-speed space connectivity, linking space probes, satellites, aircraft, UAVs, and ground stations into a space IoT for classical and quantum applications. (b) Detail of the previous image: FSO downlink transmission between NASA GRC's aircraft and a ground terminal at the GRC ACF building.

An air-to-ground quantum FSO communications system based on entangled photons is currently under development at the new Aerospace Communications Facility (ACF) of NASA's GRC in the context of NASA's Convergent Aeronautics Solutions (CAS) Quantum project. The schematic of the experimental setup is shown in Fig. 1 (b). At the transmitter, which will be eventually airborne, polarization-entangled photon pairs are created via spontaneous parametric down-conversion (SPDC). Light from the SPDC source propagates through a single-mode fiber (SMF) inside the aircraft before it reaches the transmitter telescope. Conversely, after the collection of light using an optical telescope at the ground station, light propagates from the roof to the first floor of NASA's GRC ACF building using an optical fiber that can be either single-mode, few-mode, or multimode.

Impact of optical turbulence

As a consequence of the nonuniform heating of the Earth's surface by the Sun, there are temperature-induced inhomogeneities of the refractive index of the air (Fig. 2). These inhomogeneities change over time due to wind velocity and air convection. The final result is the formation of constantly varying air cells, also called eddies, of different sizes and of different refractive indices. Depending on their sizes, these eddies can either refract or scatter various parts of the propagating beam. Due to refraction and pointing errors, the beam drifts in and out of the collecting aperture of the optical system. Scattering causes constructive or destructive interference in the plane of incidence, forming a speckled and irregularly-shaped intensity pattern at the detector. These effects can have devastating consequences on the performance of optical communications systems. While turbulence effects diminish at higher altitudes, they are a major challenge for downlink communications, such as aircraft-to-ground or satellite-to-ground links (Fig. 1(b)).

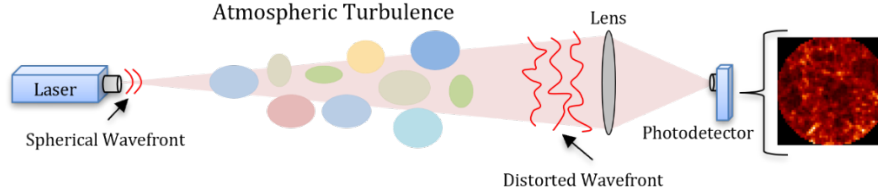


Fig. 2 Refractive index inhomogeneities of various sizes, depicted here as colorful bubbles (atmospheric turbulence eddies), cause phase distortions in the wavefronts of the received optical waves. Here, we show an example of a horizontal terrestrial link. A lens focuses the distorted wavefront onto a photodetector. Due to the random spatial phase variations of the incoming wavefront, interference occurs in the detector plane and produces speckle. As the composition of the atmosphere changes, there is a constant wander in the beam direction and the beam falls in and out of the cross section of the receiver. There is also a change in the spot size and shape, causing random fluctuations in the received power called scintillation.

Compensation of the optical turbulence-induced fading in classical communications

For classical communications, it is possible to use adaptive optical or electronic compensation schemes to invert the transfer matrix of free-space optical links and counteract scintillation.

Optical compensation methods, such as adaptive optics (AO) based on deformable mirrors or spatial light modulators (SLMs), are crucial for dynamic beam shaping and real-time correction of the random phase distortions of optical wavefronts caused by atmospheric turbulence.

An AO system includes three basic elements: a wavefront sensor, a compensator (i.e., a deformable mirror or an SLM), and a feedback scheme controlling the compensator [13], [14]. Deformable mirrors consist of a mirror membrane supported by an underlying actuator array. Each actuator can be individually deflected by electrostatic actuation to achieve the desired pattern of deformation. SLMs are electro-optical devices that can dynamically modulate the phase of an optical wavefront with high spatial resolution. SLMs use liquid crystal or microelectromechanical system (MEMS) technology to achieve wavefront control without mechanical movement as in deformable mirrors [15][15].

Electronic compensation methods of optical turbulence rely on digital signal processing (DSP) after the detection of the optical beam. The advantage is that it is possible to reduce or eliminate signal degradation without requiring mechanical or optical adaptive components.

Two notable electronic compensation methods that were proposed in the recent literature involve a) aperture averaging using incoherent combining using a multimode, optically-preamplified, FSO direct-detection receiver [16]; and b) coherent combining in the digital domain after using multiple single-mode coherent receivers to sample the distorted wavefront on different points [17].

The latter method can also enable optical Multiple-Input Multiple-Output (MIMO) FSO communications [18], where two network nodes can communicate using parallel arrays of spatially separated optical transmitters and receivers by taking advantage of the spatial diversity of different free-space propagation modes.

Coherent combining using programmable PICs

Programmable Photonics were recently proposed as adaptive optical compensators in FSO communications systems and networks, to be used independently or in conjunction with the aforementioned adaptive optical or electronic compensation schemes [19], [20].

For instance, Martinez et al. [20] used a silicon photonic chip hosting a 2D Optical Antenna Array and a self-adaptive analog programmable optical processor made of a mesh of tunable MZIs. The photonic chip acted as an adaptive coherent combiner to couple turbulent FSO beams to single-mode guided optics and add them in phase.

In addition, the French company CAILabs developed a multimode FSO receiver composed of a multi-plane light conversion (MPLC) module that takes a FSO wavefront for input and returns its projection on a 15 Hermite-Gauss (HG) mode basis, on 15 separated SMF, followed by a coherent combining programmable photonic integrated circuit. The PIC is divided into 14 unit cells, each cell being composed of two inputs (with a phase shifter in one of them), a MZI between the two inputs (with a phase shifter in one arm) and two outputs [21].

1.3 Proposed solution: Optical compensation and mode demultiplexing using programmable PICs

For FSO quantum optical satellite-to-ground or aircraft-to-ground links, we propose using programmable PICs to dynamically counteract random variations in the amplitude and phase of the superposition of received free-space propagation modes and improve signal fidelity. This technique can be used for optical compensation of optical turbulence, but also for modal dispersion and birefringence/polarization mode dispersion in the transmitter and receiver fibers.

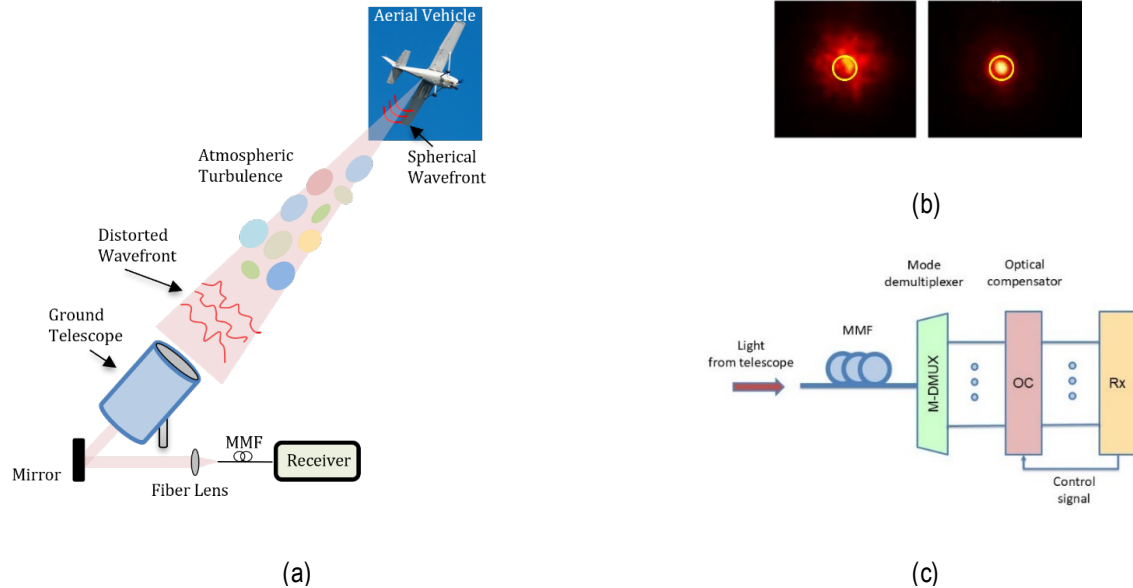


Fig. 3 (a) Illustration of laser beam propagation from an aircraft through turbulent atmosphere toward a ground-based telescope, where light is collected and coupled into either a single-mode, a few-mode, or a multimode fiber (MMF). The propagating spatial and polarization modes within the fiber are subsequently demultiplexed; (b) Speckle pattern depicted together with the fiber input cross-section (yellow circle) before and after deformable-mirror-based AO in the telescope; (c) Free-space optical (FSO) receiver subsystem including an optical compensator (OC) based on a programmable photonic integrated circuit. The feedback provided by a classical receiver (Rx) is used for real-time control of the adjustable parameters of the optical OC.

Consider an on-board FSO communication terminal capable of transmitting both classical and quantum laser communication (lasercom) signals (Fig. 3(a)). We assume that there is a dedicated system for continuous optical tracking, integrated with tip/tilt mirrors at both the transmitter and receiver, to compensate for pointing errors. We focus on the special case where classical and quantum signals are time-division multiplexed on the same wavelength. The time axis is divided into frames. Each frame is composed of classical pilot signals and qubits. Classical pilot signals are used at the beginning of each frame to characterize the transmission channel and provide channel state information at the receiver. Assuming ideal telescopes at both the transmitter and receiver, transmission impairments are introduced along the communication channel, at the single-mode fiber segment at the transmitter, the free-space propagation segment affected by atmospheric turbulence, and the fiber segment at the receiver, which may be single-mode, few-mode, or multimode. Few-mode and multimode fibers are advantageous as they capture more light from the receiver telescope compared to single-mode fibers. However, they also introduce modal dispersion and coupling among the optical modes propagating along the fiber. Single-mode fibers, on the other hand, being birefringent, scramble the states of polarization.

Let's take a closer look at the receiver subsystem shown in Fig. 3(b). The latter is composed of the receiver few-mode or multimode fiber, a coarse mode demultiplexer in the form of a photonic lantern [22], an optical compensator that can resolve and recombine spatial and polarization modes, and optical receivers that generate control signals for adjusting the free parameters of the optical compensator. Let's consider first light propagation in the transmitter SMF: The state of polarization (SOP) of light continuously changes as it travels through an SMF. Practical SMFs are slightly birefringent due to intrinsic and extrinsic mechanisms that lift the degeneracy between the x- and y- SOPs [23]. In addition, the birefringence axes orientation and the birefringence magnitude vary along the fiber. As a result, the x- and y- SOPs couple at random coupling points and travel at slightly different phases and group velocities at various fiber segments. Since the fiber changes over time due to environmental vibrations and temperature gradients along the fiber, there are ms-scale random fluctuations of the SOP at the fiber output.

Let's consider now light propagation in the receiver fiber components: Modal effects in few-mode fibers (FMFs) and multimode fibers (MMFs) are a generalization of polarization effects in SMFs [24]. There is modal birefringence since different spatial modes travel with different phases and group velocities. In addition, there is mode coupling at random points along the fiber. Furthermore, there is mode-dependent loss, i.e., higher-order modes experience higher attenuation, a feature that will be more prominent for long FMFs/MMFs. Finally, due to atmospheric turbulence and pointing errors, the illumination of the cross-section of the fiber will vary over time. This means that the number of guided modes that will be excited, as well as their amplitudes and phases, will vary over time as the input beam spot size, position, and shape will change.

If both the transmitter and the receiver contained only SMFs, a polarization controller composed of phase shifters and a MZI, could be used for polarization tracking and compensation (Fig. 4 (a)). This two-stage device is based on a concatenation of two phase waveplates with orthogonal fast eigenaxes in Stokes space [25], [26]. It can change the received state of polarization to any desired state using two consecutive rotations on two orthogonal great circles of the Poincaré sphere (PS) (Fig. 4 (b)). There are recent studies of polarization compensation in SMF-based quantum networks, e.g., see [27] and a recent field experiment performed by Qunnect [28] using such optical devices.

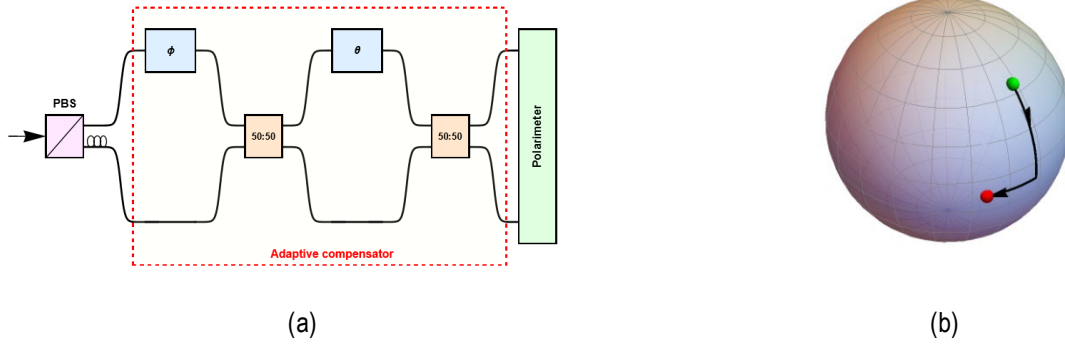


Fig. 4 (a) A schematic of a two-stage polarization controller suitable for implementation on a programmable PIC; (b) An example of a Poincaré sphere representation of the received state of polarization (green dot) and its transformation to the desired state of polarization (red dot) using two consecutive rotations on two orthogonal great circles of the Poincaré sphere.

In the more realistic case where the transmitter and the receiver contain both SMFs and FMFs/MMFs, the overall channel transfer matrix has dimensions $N \times N$, where N is the number of spatial and polarization modes. A proof of principle that the optical transfer matrix of very multimode optical fibers can be measured and inverted is given by [29]. If channel transfer matrix inversion is performed successfully, the SOP at the output of the receiver has high fidelity to the transmitted entangled photon SOP.

We will focus on programmable PICs that can invert any $N \times N$ fiber transfer matrix. Assuming that, to first approximation, mode-dependent loss is negligible for short few-mode and multimode fibers, the transfer matrix of the fiber is unitary. Therefore, we are looking into candidate PICs architectures that can be used for realization of arbitrary $N \times N$ unitary matrices. For this purpose, several silicon photonics architectures based on 2D arrays of phase shifters MZI, have been proposed in the literature: The triangular mesh [30], the rectangular mesh [31], the diamond mesh [32], the Bokun mesh [33] etc. An example of arbitrary 20×20 unitary matrices using (a) a triangular mesh [23]; (b) a rectangular mesh [25] is shown in Fig. 5.

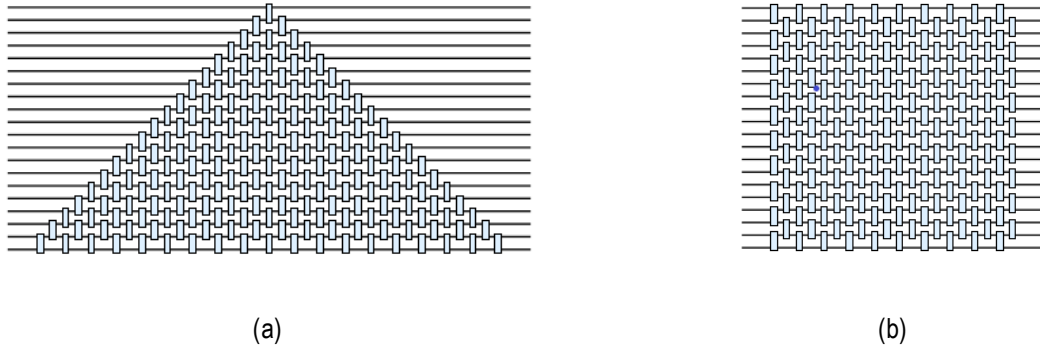


Fig. 5 Realization of arbitrary 20×20 unitary matrices using (a) a triangular mesh [30]; (b) a rectangular mesh [31]. Lightblue rectangles represent the fundamental building block shown in Fig. 4(a).

1.4 Technical Task Descriptions

In this section, we present a comprehensive overview of each task, outlining its objectives, key focus areas, expected outcomes, and success metrics to track progress throughout the project's duration.

Task 1: Design of resilient and highly-scalable optical compensators based on programmable PICs and multimode polarimeters (Lead: Prof. I. Roudas)

The primary focus of this research task will be on designing, modeling, and simulating scalable, fault-tolerant, and fast adaptive optical compensators in silicon photonics, based on the simulation setup of Fig. 6. This task

focuses on the optimal design of the three major modules needed for compensation, namely the programmable photonic processor, the optical monitor (i.e., multimode polarimeter), and the control electronics.

For the architecture of the programmable photonic processor, we will use as our starting point the rectangular mesh [31], shown in Fig. 5(b), which is more compact and more tolerant to fabrication errors compared to the one in Fig. 5(a). Several variants of the rectangular mesh architecture have been proposed in the recent literature [34], [35], [36] to enhance device reliability by improving tolerance to fabrication errors through the introduction of redundancy. Our ultimate goal is to identify the best out of the proposed architectures for photonic processors [34], [35], [36] and optimize it in order to be able to handle up to 1,000 spatial and polarization modes, which are necessary for large diameter MMFs.

Our team recently proposed and studied several novel architectures of multimode polarimeters based on direct detection, which can infer all N^2 Stokes parameters using only a small number of photodiodes on the order of $O(N)$ [37]. In particular, our group invented the low-hardware-complexity multimode polarimeter shown in Fig. 7, where the number of photodiodes is only $3N - 2$ [38]. This multimode polarimeter uses an asymmetric variant of the common 90-degree optical hybrid as a building block to create pairwise superpositions of the constituents of the received multimode electric field (Fig. 7 (a)). The general layout of the multimode polarimeter for any even N is shown in Fig. 7(b). The architecture is composed of two successive layers of interconnected asymmetric optical hybrids, i.e., $N/2$ input optical hybrids (magenta) and $N/2 - 1$ output optical hybrids (green), respectively. This design results in a total of $N - 1$ asymmetric optical hybrids and only $3N - 2$ photodiodes. Each optical hybrid (Fig. 7 (a)) consists of two asymmetric directional 2x2 couplers, two directional 2x2 3-dB couplers, a 90-degree phase shifter, and a fiber permutation between stages. The splitting ratio of the asymmetric couplers needs to be optimized in order to minimize the noise impact in different regimes.

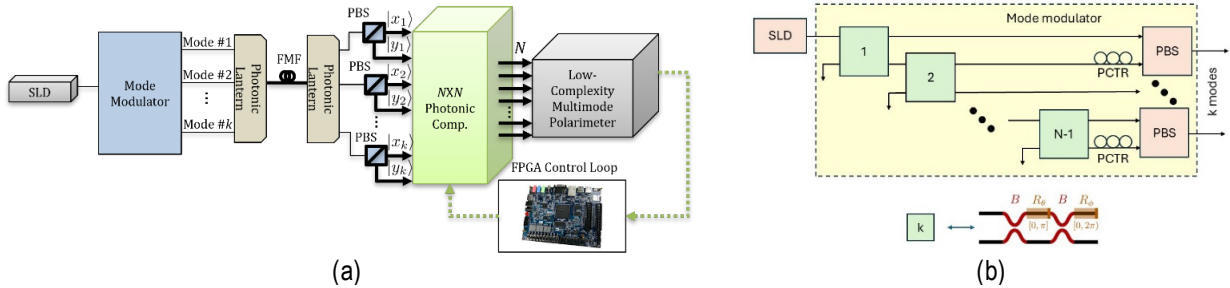


Fig. 6 Simulation block diagram of an $N \times N$ optical compensator acting on k spatial mode groups with two states of polarization each, i.e., $N=2k$. Light from a semiconductor laser diode (SLD) is evenly distributed among k dual-polarization IQ modulators to generate any superposition of $N=2k$ spatial and polarization degrees of freedom (DoFs). Photonic lanterns are used for mode multiplexing at the transmitter and coarse demultiplexing at the receiver. On the receiver side, after the photonic lantern, the polarization components of the individual modes are separated and fed to the programmable photonic processor. The channel state is monitored by a multimode polarimeter. The measurements of the polarimeter are used by a control algorithm to iteratively adjust the free parameters of the programmable photonic processor (i.e., the voltages of the phase shifters to counteract modal effects in the FMF).

Lastly, we will implement fast control algorithms to adjust parameters of large programmable photonic processor on μs timescales. We will use the measured state of polarization (SOP) provided by the multimode polarimeter to assess whether the transmission channel is fully inverted or not and decide whether the parameters of the programmable photonic processor need any further adjustments [39]. In this research activity of Task 1, we will compare various adaptive equalization algorithms, either blind or data-aided, such as the least mean square (LMS) [40], the independent component analysis (ICA) [41], and the Kalman filter [42] and its variants, e.g., the extended Kalman filter (EKF) [43]. The aforementioned algorithms will be ranked based on their performance in terms of residual modal and polarization crosstalk and convergence speed.

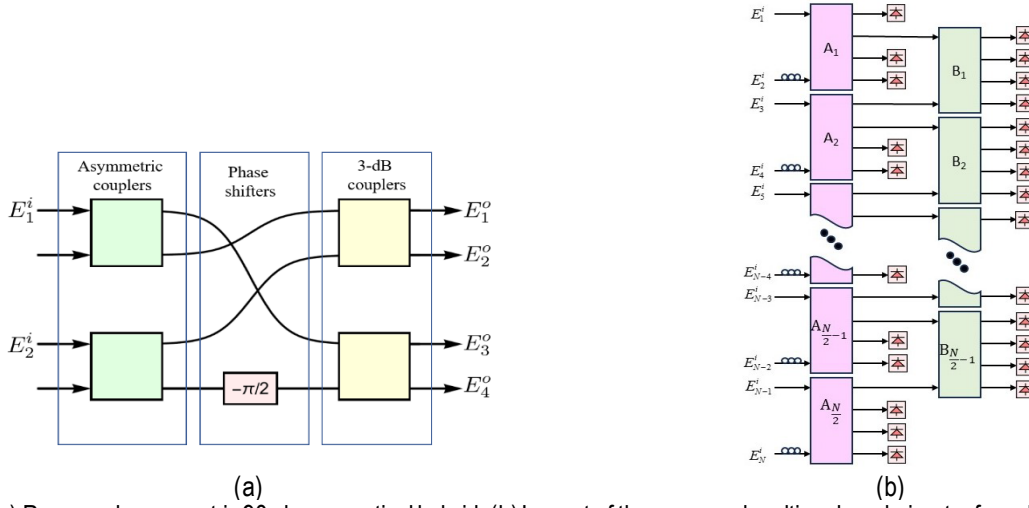


Fig. 7 (a) Proposed asymmetric 90-degree optical hybrid; (b) Layout of the proposed multimode polarimeter for arbitrary even number of spatial and polarization degrees of freedom N , composed of $N-1$ cascaded asymmetrical 90-degree optical hybrids.

Task 2: Experimental FSO communication systems (Leads: Prof. K. Repasky and Prof. B. LaMeris)

We intend to build a tabletop experiment (Fig. 8) to test an adaptive optical polarization compensator that can counteract rotations in the state of polarization of the transmitted signal caused by the birefringence of the optical fibers in the transmitter and the receiver. The setup uses free space optical components.

The heart of the setup is a two-stage automated polarization controller like the one shown in Fig. 4(a), composed of a free-space MZI and two bulk, MHz-bandwidth, LiNbO₃ phase modulators as tunable phase shifters (Fig. 8). The first stage of the automated polarization controller rotates the incoming state of polarization along the x-axis on the PS and the second stage rotates the state of polarization along the y-axis on the PS.

We will build two copies of the automated polarization controller using one as a polarization modulator at the transmitter to create any desired SOP and the other as an adaptive optical polarization compensator at the receiver. The setup also includes a polarization scrambler, a cost-efficient polarimeter (Fig. 9), and an array of FPGA cards in terms of control electronics (Fig. 8).

In the diagram shown in Fig. 8, at the transmitter, located on the left, we use a continuous-wave (CW) laser and a two-stage automated polarization controller, which functions as a polarization modulator. This setup can generate any desired polarization state by appropriately adjusting the phase modulators (in red). These polarization states can be used as pilot symbols to estimate the channel transfer matrix at the receiver. The beam then passes through a polarization scrambler with fiber pigtails.

The polarization scrambler randomizes the instantaneous state of polarization of the output to emulate the time-varying-birefringence of the transmitter and receiver SMFs. The polarization scrambler consists of a series of fiber squeezers oriented at 45 degrees from each other that press on the fiber at various points to induce birefringence due to the photo-elastic effect. The fiber squeezers are driven independently with different modulation frequencies to ensure full coverage of the PS within milliseconds. However, even without the scrambler, random mechanical deformations of the optical fibers caused by building vibrations lead to continuous fluctuations in the received SOP of the optical wave on the PS, though full coverage of the sphere is not achieved since the laboratory environment is relatively stable.

The output from the scrambler is directed to a polarization beam splitter (PBS), which decomposes the signal into its x and y components and feeds them into the two ports of the optical adaptive polarization equalizer. This equalizer is an automated polarization controller identical to the one used at the transmitter. Ideally, by properly

adjusting the two phase modulators, the polarization controller inverts the channel transfer matrix, where the polarization scrambler acts as the channel.

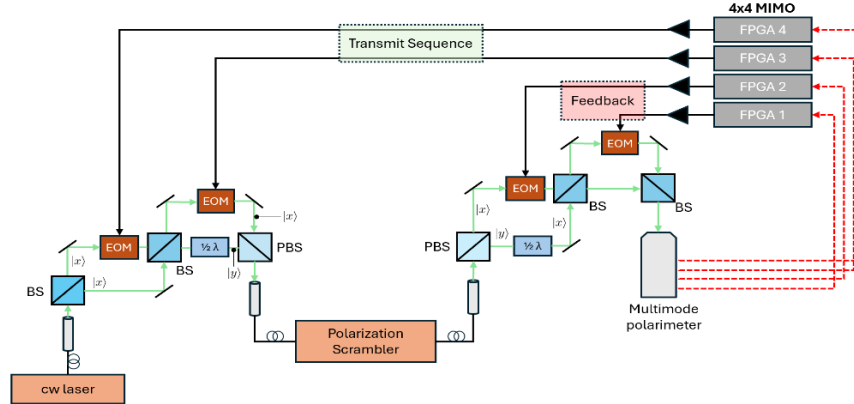


Fig. 8 Experimental setup to test the 2x2 optical compensator. A polarization scrambler will be used to scramble the transmitted SOP on a time scale of milliseconds. FPGA cards will be used to dynamically control the modulator's and the 2x2 compensator's phase shifters at microsecond speeds, allocating sufficient control time for convergence.

To verify whether the phase modulator settings of the adaptive polarization equalizer are correct, the equalizer outputs are fed into a polarimeter. For fully polarized light, detecting the Stokes vector requires only four measurements [44]. Therefore, the polarimeter can be designed with just four single-ended photodiodes that detect four different superposition states of the x and y components of the received polarization state. To generate the appropriate superposition states of the x and y components of the received polarization state, we first turn y into x and then use an asymmetric optical hybrid of Fig. 7(a), implemented with FSO components as shown in Fig. 9(a). We have demonstrated through analysis and simulation that, under thermal noise conditions, the proposed polarimeter outperforms alternative six-photodiode polarimeter topologies found in literature in terms of sensitivity [38].

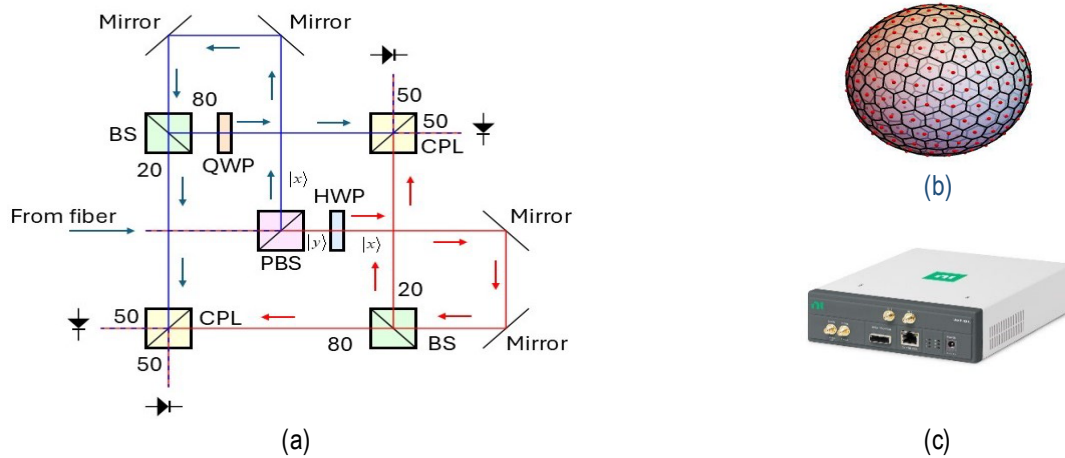


Fig. 9 (a) Block diagram of a cost-effective polarimeter using only four photodiodes (denoted by the grey polygon in the experimental setup in Fig. 8). It is based on the asymmetric 90-degree optical hybrid in Fig. 7(a) and is implemented here using FSO components; (b) Optimized constellation of 256 pilot symbols that are equidistant on the Poincaré sphere; (c) Ettus USRP N210 FPGA card.

The experimental setup in Fig. 8 operates at a relatively slow rate due to the mechanical nature of the polarization scrambler. FPGAs are commonly used in software-defined communications systems for digital signal processing. An array of synchronized commercially-available Software-Defined Radio (SDR) transceivers, such

as the USRP N210 platforms from Ettus Research (Fig. 9(c)), can be used in the feedback loop to dynamically adjust the phase shifters of the transmitter's polarization modulator and the adaptive polarization equalizer in real-time. These platforms are based on the Xilinx® Spartan® 3A-DSP 3400 FPGA and are equipped with 100 MS/s dual ADC and DAC. They can be programmed using a graphic user interface such as MATLAB Simulink, GNU Radio, or LabView.

We can use a block of M pilot symbols $M > N$ (Fig. 9(b)) that we transmit at the beginning of each frame to compute the pseudo-inverse of the channel matrix based on the corresponding measurements of the received Stokes vectors from the polarimeter [39]. Alternatively, we can perform blind adaptive channel equalization, where the control circuit compares the received Stokes vector with the expected one and generates an error signal. The latter is used by a recursive control algorithm, such as LMS, Kalman, or ICA, to adjust appropriately the phase modulator settings of the adaptive polarization equalizer.

The reconfiguration cycle of the programmable PIC should last on the order of a few μ s, to allow enough time to transmit quantum signals between reconfiguration cycles. Finally, the software layer will include auxiliary functions necessary to maintain chip temperature stability, to set the DC bias voltages of the MZI modulators, etc.

Task 3: Quantum applications of programmable photonics (Lead: Prof. M. Jaffe)

Apart from optical compensation, a major goal of the proposed project is to study programmable PIC architectures implementing $N \times N$ unitary matrices for quantum applications [45].

Photonic qubits are essential for quantum networking. Long distance fiber-based networks will also require interfacing to intermediary qubits in the form of quantum repeaters [46]. Since photons do not interact with each other, they cannot straightforwardly enact two-qubit gates. While photonic, "measurement-based," quantum computation [47] is being pursued by some [48], [49], thus far stringent technical obstacles, such as loss, have limited photonic quantum information processing below the gate and circuit performances demonstrated on qubit platforms such as superconducting circuits [50], trapped ions [51], and neutral atoms [52].

Photonic interconnects between quantum systems [53], [54], [55] are thus a near-certain requirement of utility-scale quantum systems regardless of platform. However, effects such as atmospheric turbulence and polarization drift in optical fibers can rotate the qubit's Hilbert space, scrambling the quantum information it encodes. It is thus critical to counteract these rotations. A second hurdle arises in connecting arrays of qubits, either in a quantum memory or computational array. Due to the large number of operations expected to be required for useful quantum computations [56], [57], it is likely that a useful quantum computer will consist of several or many modules with entanglement distribution channels between them. Photons represent an obvious choice to mediate such entanglement distribution, but the challenge of routing one $N \times N$ qubit array to another $N \times N$ qubit array is nontrivial. While fixed connectivity represents one path forward (which still requires the aforementioned noisy channel correction), arbitrary connectivity would enable a wider class of quantum algorithms and error correction codes [58], [59]. Architectures of $N \times N$ programmable-unitary PICs as the ones shown in Fig. 5 could interconnect many modular arrays of photon-coupled qubits.

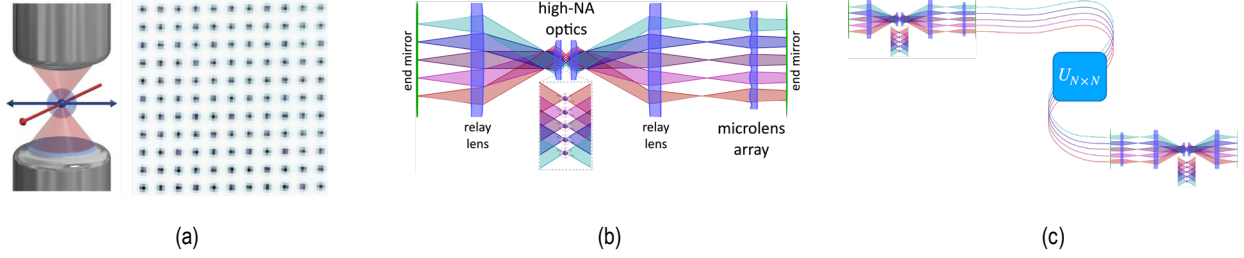


Fig. 10 (a) Neutral atom qubit array [32] with each qubit in its own optical tweezer; (b) Optical design for building a cavity tweezer array, providing strong coupling between each neutral atom qubit and its own cavity mode; (c) Interfacing two such systems for distributed quantum information. The PIC-based arbitrary unitary matrix $U_{N \times N}$ allows for any-to-any qubit coupling between the two systems, as well as correction of environmental noise in the fiber channels.

Co-I Matt Jaffe's lab at Montana State is working on coupling such qubit arrays to each other. Building on his prior work [60], single neutral atom qubits can be strongly coupled to a bulk optical cavity at low finesse by use of intracavity lenses to generate small (sub- μm) cavity waists. By introducing a microlens array, a cavity tweezer array can be realized, constituting an array of qubits each coupled to its own optical cavity (Fig. 10b).

The strong photon-qubit coupling enables high-rate, heralded, high-fidelity generation of Bell pairs between two such systems with a photonic link between them. The qubit phase must be stabilized over this photonic link, which the PIC platform proposed here will accomplish. By linking separate cavity-coupled qubit *arrays* (rather than single qubits in single cavities) as in Fig. 10(c), the arbitrary $N \times N$ unitary matrix of the programmable PICs proposed here enable coupling between qubit arrays with arbitrary (unitary) mapping between qubit sites. This all-to-all connectivity greatly expands the classes [61] of quantum algorithms and quantum error correction codes that can be used for quantum computation when combining modules. It even allows for rapidly producing photon-mediated Bell pairs in parallel between arbitrary qubit pairs *within a single computational module* as entanglement resources to be consumed by subsequent computation using local gates.

Beyond the above applications, photonic links can also interface with the small-waist cavities of the Jaffe lab to form a multiplexed quantum memory from solid state qubits such as rare-earth ions [62]. This would find application not only in quantum information storage [63] but potentially in use as a quantum repeater, or even for flying-optical-photon to microwave photon transduction [64] for interfacing to superconducting qubits.

Task 4: Fabrication and experimental characterization of various PIC designs (Lead: NASA GRC)

This task focuses on manufacturing and characterizing prototypes of optical compensators and quantum optical devices developed in Tasks 1 and 3. Mr. B. Vyhalek from NASA GRC has been creating proof-of-concept programmable PIC prototypes. An initial prototype, a 4×1 mode multiplexer with a reconfigurable core of 3 Programmable Unit Cells in a triangular layout, was fabricated in silicon photonics at AIM Photonics (Fig. 11). Of particular interest to the CAS quantum project is to continue and expand this early work to more sophisticated programmable PICs that are based on [34], [35] with possible improvements, e.g., [36].

MSU will collaborate with the Optics and Photonics Branch at NASA GRC to design the architecture of a large-port-count, highly scalable, fault-tolerant, photonic processor to be used as an adaptive optical compensator for fast polarization tracking and stabilization in multimode quantum networks for entanglement distribution. The group at NASA GRC will develop the full electronic and software stack for the fabrication of the chip using the AIM Photonics Process Design Kit (PDK), will fabricate the chip at AIM Photonics, and will test its performance in order to provide technical feedback for the Tasks 1-3 above.



Fig. 11 (a) Schematic of a 4×1 mode multiplexer; (b) Prototype 4×1 self-regulating PIC designed at NASA GRC and fabricated at AIM Photonics. Monitoring photodiodes are incorporated in the output arms of all individual MZI interferometers. Dummy MZIs are also included to equalize the propagation loss in all paths.

1.5 Early results

Two small-scale preliminary studies, representing different critical research activities within the proposal, were conducted by the Sci-I, the co-Is and the GRAs during the period between the preproposal and final proposal submissions.

1. Impact of atmospheric turbulence on terrestrial FSO links

The first study examined the impact of atmospheric turbulence on terrestrial FSO links. Our team performed experimental measurements of the refractive index structure parameter of a horizontal terrestrial link throughout an entire day using a scintillometer. Through analysis and simulation, we estimated the average error probability for various modulation schemes based on a statistical model of the atmospheric turbulence that calculates the probability density function (pdf) of the received SNR based on the refractive index structure parameter. For instance, Fig. 12 shows the estimated daily evolution of the average bit error rate of differential binary phase shift keying (DBPSK) (in blue) plotted against the refractive index structure parameter (in orange). We notice that, in the middle of the day when there is strong turbulence due to the heating of the ground from the Sun the performance of optical links greatly deteriorates.

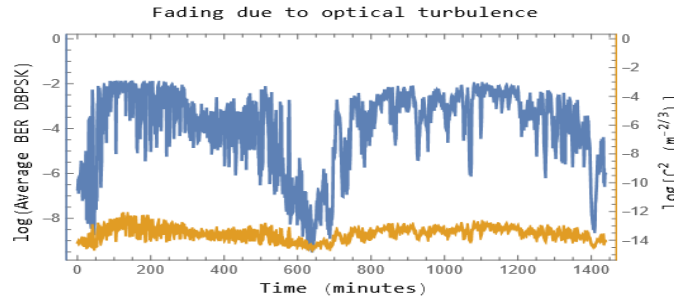


Fig. 12 Daily evolution of the DBPSK bit error rate (in blue) plotted together with the refractive index structure parameter (in orange).

2. Simulation of a programmable photonic integrated circuit with a 4×4 unitary transfer matrix

The second study involved the simulation of a programmable integrated circuit featuring a 4×4 unitary transfer matrix, based on a triangular mesh architecture of MZIs and phase shifters proposed by Reck et al. [30]. We used two different commercially-available software tools to gain a deeper understanding of various performance characteristics of the circuit.

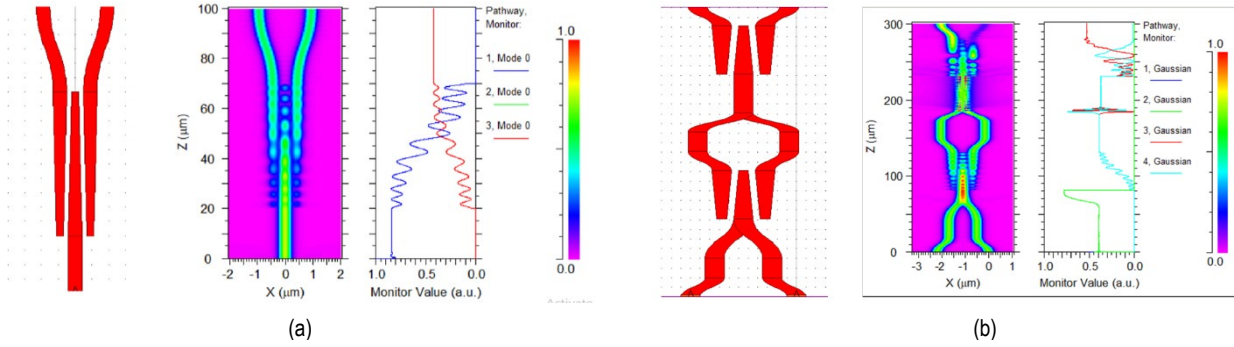


Fig. 13 EM wave propagation inside (a) a 3-dB splitter. (b) a Mach-Zehnder interferometer, which is the basic building block of the universal multiport interferometer architectures proposed by Reck et al. and Clements et al.

Initially, we studied electromagnetic wave propagation inside a 3-dB splitter and a MZI using the finite-difference time-domain (FDTD) method (see Fig. 13). These simulations enable the calculation of the impact of the geometric characteristics of various components of the programmable PIC on its performance and accounts for potential fabrication imperfections. Subsequently, we studied waveform propagation in the 4×4 Reck triangular mesh architecture using an optical communications systems simulation tool (Synopsys RSoft OptSim, Fig. 14). We programmed the phase shifters within the programmable PIC to separate various orthogonal mode combinations to different outputs. Waveform level simulations of various adaptive optical compensator architectures could be used to understand their resilience vis-a-vis to different fabrication imperfections. For instance, using this simulation tool, we can investigate how the propagation losses, splitting ratio variations, and phase setting imperfections may accumulate in large architectures.

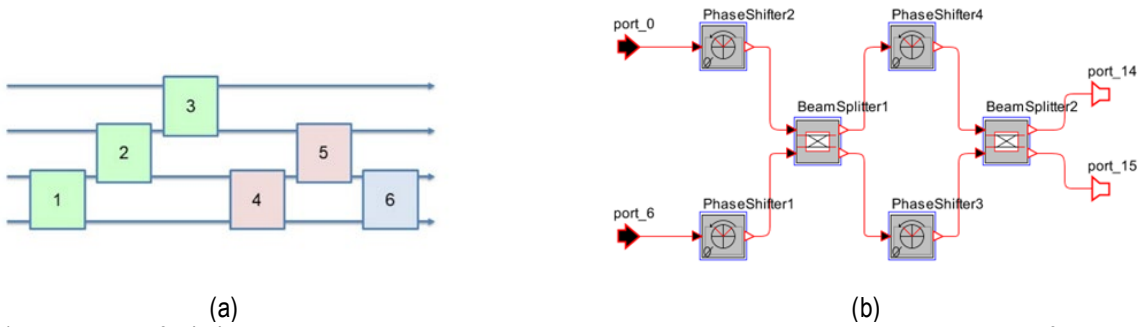


Fig. 14 (a) Architecture of a 4×4 programmable photonic integrated circuit based on a triangular mesh architecture of Mach-Zehnder interferometers and phase shifters proposed by Reck et al.; (b) Module in the graphic user interface of Synopsys RSoft OptSim representing the fundamental building block of the programmable photonic integrated circuit composed of a Mach-Zehnder interferometers and two phase shifters.

1.6 Intrinsic Merit

One of the major challenges for future NASA missions is the development of a high-capacity, free-space optical network, which will be an extension of the terrestrial fiber-optic network in the sky. The free-space optical network must be capable of providing nearly-continuous connectivity among classical and quantum devices. However, optical signals from satellites to Earth experience distortion due to atmospheric turbulence, that causes intensity fluctuations (scintillation) and signal fading. For this purpose, in this project, we propose to design and demonstrate programmable photonic integrated circuits that can be used as adaptive equalizers to correct the wavefront phase distortions acquired during the propagation of lightwaves through a turbulent atmosphere.

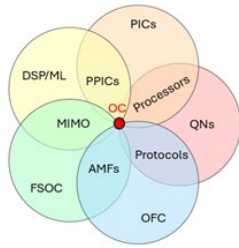


Fig. 15 Venn diagram illustrating that adaptive optical equalizers (OC) based on programmable photonic integrated circuits (PPICs) lie at the cross-section of several important technological areas, namely photonic integrated circuits (PICs), optical fiber communications (OFC), free-space optical communications (FSOC), quantum networking (QN), and digital signal processing/machine learning (DSP/ML). All these areas can benefit from the development and commercialization of adaptive optical equalizers and hold the promise of fast growth in the next decade and beyond.

As illustrated in the Venn diagram in Fig. 15, adaptive optical equalizers based on programmable photonic integrated circuits (PPICs) are positioned at the intersection of multiple key technological domains, some of which are forecasted to experience a compound annual growth rate (CAGR) up to 35% in the next years. Therefore, we anticipate that the knowledge gained from this project will enable the Optical Communications Group at Montana State University to successfully participate in future non-EPSCoR research projects in these adjacent areas and attract extramural funding both from government agencies and the private sector.

The Optical Communications Group at Montana State University has investigated over the years a large number of cutting-edge research problems in point-to-point optical communications systems and networks, including emerging photonic device technologies for optical communications. It has significant expertise in transmission effects occurring in fiber-optic and free-space optical networks, as well as their compensation using digital signal processing at the optical transceivers, for various multidimensional modulation formats.

In this project, an interdisciplinary team from Montana State University and NASA GRC has come together to explore ways to optimize the design of adaptive optical equalizers based on programmable photonic integrated circuits. Our major goal is the accurate modeling, simulation, and experimental characterization of all parts of the adaptive optical equalizer, which is comprised of a resilient and highly-scalable photonic processor, a multimode polarimeter, and a real-time control circuit. Innovations unique to this project are the proposed high-speed low-complexity multimode polarimeter (Fig. 7 and Fig. 9), the use of pilot signals based on equidistant points on the PS [40], and the use of multiple real-time control algorithms to counteract the mode mixing in the channel and their implementation on FPGAs [39]. A more forward-looking, innovative aspect of this project is the investigation of quantum applications of programmable photonics as routers between $N \times N$ neutral atom qubit arrays.

The proposed research project builds on existing infrastructure initiatives of Montana State University: First, it is entirely within the scope of Montana State University's Applied Quantum CORE program. In addition, Montana State University has heavily invested in the miniaturization of optical components through the MonArk NSF Quantum Foundry and the Montana Microfabrication Facility (MMF). Furthermore, the project is aligned with the activities of the Optical Technology Center (OpTeC) of Montana State University. Note that both MSU and the State of Montana have identified Optics as one of its key pillars for economic growth.

Due to the state's business-friendly environment there are several photonics startups near the campus. MSU has close ties with the local industry providing research ideas and a skilled workforce. The Montana Photonics and Quantum Industry Alliance also links optics, photonics, and quantum businesses with local universities. MSU's Applied Quantum CORE (QCORE), supported by the Air Force Research Laboratory, advances quantum technologies from research to market. In addition, Montana's Headwaters \$41 million Tech Hub advances smart photonic sensing. All the above provide opportunities for the commercialization of the intellectual property developed during the project on programmable PICs by local optics and photonics companies and can provide employment opportunities for the GRAs after their graduation.

1.7 NASA Alignment and Partnerships

The proposed topic lies within NASA's Research Areas of Interest for FY2025, see Appendix A: NASA Mission Directorates and Center Alignment, Section A: Basic Research Topics, Subsection A.6.3 Glenn Research Center (GRC), Topic: Quantum Sensors, Communications, and Networks.

NASA's Glenn Research Center (GRC) Optics and Photonics Branch is involved in all aforementioned research topics. Apart from the leader of the Quantum Communications Group at NASA's GRC Dr. J. Lekki, we will collaborate with NASA GRC Research Electronics Engineers Yousef Chahine and Brian Vyhnaelek. They will provide technical support for various research activities of the program, such as specifications for PPICs, which form the basis of our proposal.

The PI of this project visited NASA GRC Optics and Photonics Branch for 11 weeks during his sabbatical leave from Montana State University for the 2023-2024 academic year and was involved in designing a configurable programmable PIC for applications in FSO links. The proposed collaboration is a continuation of this research activity aiming to establish a permanent partnership between the two institutions.

Quantum Science & Technology, Integrated Photonics, and FSO communications are key priorities for NASA. Relevant topics in the solicitation include Quantum coherence, entanglement, and precision measurements (SMD); Quantum computing, sensors, and algorithms (ARC); Integrated photonics, quantum networking (GSFC); and Optical communications and navigation (JPL). The proposed project aligns with these priorities, creating opportunities for future partnerships between MSU, NASA Centers, and JPL.

This project directly supports Montana NASA EPSCoR's goals by fostering collaboration between MSU and NASA Glenn Research Center (GRC) in designing and testing PPICs for FSO communications. It contributes to NASA's mission in Quantum Science, Integrated Photonics, and FSO communications while leveraging Montana's existing infrastructure. Additionally, the project integrates with Montana's Quantum Workforce Development Initiative, ensuring student training in quantum technologies and strengthening ties to local industry through the Montana Photonics and Quantum Industry Alliance. Successful completion will position MSU to compete for national research projects in quantum communications.

Finally, the technologies studied in this proposal (i.e., FSO communications, programmable photonics, adaptive optics using programmable PICs) align well with the Strategic Framework of NASA Space Technology Mission Directorate (STMD), which focuses on enabling high-data-rate optical and quantum communication systems for near-Earth, cislunar, and lunar environments. They address the needs of the following STMD strategic areas in the near to mid future (beyond 2030):

- Near-Earth Communications: Over 200 Gbps direct-to-Earth optical downlink for small LEO satellites and 1-100+ Gbps data transfer between satellites.
- Cislunar and Lunar Communications: Coherent 1-10+ Gbps direct-to-Earth links from lunar orbit.
- Lunar surface communications: Fiber-optic links between base stations on the Moon's surface.
- Quantum Communications: Key focus areas include entanglement swapping, quantum memories, and quantum networking with quantum repeaters to extend communication range.

Integrated Photonics is ranked as a medium-priority technology in NASA's 2016 Space Technology Roadmap, with strong potential for near-Earth optical communications, intersatellite links, and applications in sensors for autonomous landing, docking, and scientific measurements.

2. Management Plan

2.1 Project Management

The project will be conducted by two multidisciplinary teams from Montana State University (MSU) and the Optics and Photonics Branch of NASA Glenn Research Center (GRC).

Montana State University Team

The MSU team comprises five senior members, three from the Department of Electrical and Computer Engineering and two from the Department of Physics:

- PI Dr. Angela Des Jardins (Associate Research Professor, Director of Montana Space Grant Consortium, Department of Physics) will provide overall guidance for the project, ensuring alignment with NASA EPSCoR goals, and overseeing project execution. She will be the interface between NASA and MSU and she will be submitting reports to NASA.
- Sci-I Dr. Ioannis Roudas (Professor, Gilhousen Telecommunications Chair, Department of Electrical and Computer Engineering) specializes in optical communications. He will be the technical lead of the project on behalf of MSU and will have ownership of Task1 focusing on the design and simulation of adaptive optical equalizers based on programmable photonic integrated circuits (PICs).
- Co-I Dr. Brock J. LaMeres (Professor, Director of Montana Engineering Education Research Center, Department of Electrical and Computer Engineering) specializes in Radiation Tolerant Computer Systems and Reconfigurable Computing. He will lead the FPGA-based control electronics development in Task 2. Dr. LaMeres research group has been developing a radiation-tolerant computer (RadPC) currently traveling to the Moon via NASA's Commercial Lunar Payload Services program. Additionally, he is the founder and president of Resilient Computing, a company dedicated to fault-tolerant computing solutions for space applications.
- Co-I Dr. Kevin Repasky (Professor, Department of Electrical and Computer Engineering) specializes in laser remote sensing, electro-optics, feedback and control, and optical communication technology. He will lead Task 2, which focuses on development of an experimental free-space optical (FSO) setup for testing a 2x2 adaptive optical equalizer.
- Co-I Dr. Matt Jaffe (Assistant Professor, Department of Physics) specializes in quantum cavity electrodynamics. He will lead Task 3, which focuses on quantum applications of programmable PICs for the interconnection of quantum memories based on neutral atom qubit arrays.

All senior MSU faculty have previously received NASA research awards.

The MSU team will also integrate four graduate research assistants (GRAs) providing them with the opportunity to conduct research in classical and quantum free-space optical communications and programmable photonic integrated circuits, all of which are emerging fields with significant potential.

NASA Glenn Research Center team

Key personnel from NASA GRC and their anticipated contributions include:

- Dr. John D. Lekki (Senior Researcher, Instrumentation and Measurement Systems, Optics and Photonics Branch): Dr. Lekki serves as the Principal Investigator for the NASA GRC Quantum Communications research area. His research group is developing technologies and capabilities to support space-based quantum communication and sensing networks and is currently focused on developing technologies for the transmission of quantum optical signals to aircraft and satellites, including the NASA Quantum Metrology Laboratory, the NASA Quantum Communications Analysis Suite, and an integrated quantum communications testbed. He will provide overall guidance for the project, ensuring alignment with the NASA GRC Quantum Communications research goals.
- Dr. Yousef K. Chahine (Research Electronics Engineer, Optics and Photonics Branch) specializes in modeling and simulation of space-based networks employing entanglement distribution. He has significant expertise in the modeling of light propagation through the atmosphere in the presence of optical turbulence. He will provide technical guidance to the MSU team for Task 1 and Task 3.
- Mr. Brian Vyhnalek (Research Electronics Engineer, Advanced High Frequency Branch, Communications & Intelligent Systems Division) has been creating proof-of-concept programmable PIC prototypes. An initial prototype, a 4×1 mode multiplexer with a reconfigurable core of 3 Programmable Unit Cells in a triangular layout, was fabricated in silicon photonics at AIM Photonics. In addition, on the same tape-out was the 7x1 coherent combiner PIC, and is currently being fabricated is a 3-mode, dual polarization, rectangular mesh PIC
- He has also worked on applications to space communications and sensing, including optical transmitter and receiver technology primarily in the areas of single-photon detectors, fiber optics, and propagation modeling. He will be the leading researcher in Task 4 of the proposal. He will provide technical guidance to the MSU team for the modeling and simulation of the programmable photonic integrated circuits in Task 1.

Project meetings

The Sci-I and Co-Is from MSU and NASA GRC's senior researchers will meet bi-weekly through Zoom to discuss the project progress, to share interim research results, and engage in the planning of future research activities. MSU GRAs involved in the project will interact on a daily basis with the corresponding Task leaders at MSU. MSU students and faculty will be eligible to visit NASA Glenn Research Center in the summers in the context of the NASA Glenn Summer Fellowship Program. Additionally, NASA GRC's senior researchers will try to periodically visit MSU.

Quarterly virtual reviews will give researchers from MSU and NASA GRC the opportunity to come together to discuss key research outcomes.

The research will culminate in joint publications in scientific journals and conferences through the collaboration of researchers among both organizations.